Microbes in Human Welfare

Besides macroscopic plants and animals, microbes are the major components of biological systems on this earth.

Where they live

- microbes are present everywhere, in soil, water, air, inside our bodies and inside those of other animals and plants
- they are present even at sites where no other life-form could possibly exist, such as deep inside the geysers, that is thermal vents, where the temperature may be as high as 100 degrees C
- they are found deep in the soil, under layers of snow several metres thick, and in highly acidic environments

What counts as a microbe

- protozoa, bacteria and fungi
- microscopic animal and plant viruses
- viroids, and also prions, which are proteinacious infectious agents

Growing them

- microbes such as bacteria and many fungi can be grown on nutritive media to form colonies that can be seen with the naked eyes
- such cultures are useful in studies on micro-organisms
- which kingdoms among the living organisms contain micro-organisms, and which are the ones that are only microscopic, is worth recalling from Class XI

In this chapter

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Rod-shaped bacterium, the bacillus form

Spherical bacterium, the coccus form

A rod-shaped bacterium carrying flagella

Figure 8.A Bacteria: (a) Rod-shaped, magnified 1500X; (b) Spherical shaped, magnified1500X; (c) A rod-shaped bacterium showing flagella, magnified 50,000X

A bacteriophage, the virus of bacteria

Adenovirus, which causes respiratory infections

Tobacco Mosaic Virus, rod-shaped

Figure 8.B Viruses: (a) A bacteriophage; (b) Adenovirus which causes respiratory infections; (c) Rod-shaped Tobacco Mosaic Virus (TMV). Magnified about 1,00,000–1,50,000X

Colonies of bacteria, grown in a petri dish

A fungal colony, grown the same way

Both large enough to be seen with the naked eyes

Figure 8.C (a) Colonies of bacteria growing in a petri dish; (b) Fungal colony growing in a petri dish

8.1 Microbes in household products

Curd

- we use microbes or products derived from them everyday, a common example being the production of curd from milk
- micro-organisms such as Lactobacillus and others, commonly called lactic acid bacteria (LAB), grow in milk and convert it to curd, producing acids as they grow that coagulate and partially digest the milk proteins
- a small amount of curd added to fresh milk as inoculum or starter contains millions of LAB, which at suitable temperatures multiply and convert the milk to curd, and this also improves its nutritional quality by increasing vitamin B12
- in our stomach too the LAB play a very beneficial role in checking disease-causing microbes

Dough, bread and traditional foods

- the dough used for making foods such as dosa and idli is also fermented by bacteria, and the puffed-up appearance of the dough is due to the production of CO_2 gas, which leaves the metabolic pathway responsible, and the source of the bacteria, to be worked out
- the dough used for making bread is similarly fermented, using baker's yeast, Saccharomyces cerevisiae
- a number of traditional drinks and foods are also made by fermentation by microbes, such as Toddy, a traditional drink of some parts of southern India made by fermenting sap from palms
- microbes are also used to ferment fish, soyabean and bamboo-shoots to make foods

Cheese

- cheese is one of the oldest food items in which microbes were used
- different varieties of cheese are known by their characteristic texture, flavour and taste, the specificity coming from the microbes used
- the large holes in Swiss cheese are due to the production of a large amount of CO_2 by a bacterium named Propionibacterium sharmanii
- Roquefort cheese is ripened by growing a specific fungi on it, which gives it a particular flavour

8.2 Microbes in industrial products

Made at industrial scale

- even in industry, microbes are used to synthesise a number of products valuable to human beings, beverages and antibiotics being some examples
- production on an industrial scale requires growing microbes in very large vessels called fermentors

Fermentors, the very large vessels

Microbes grown in them at industrial scale

Where beverages and antibiotics are made

Figure 8.D Fermentors

A fermentation plant, photographed whole

Vessels, pipework and the plant around them

The industrial setting of the same process

Figure 8.E Fermentation Plant

8.3 Fermented beverages

Yeast, and what it makes

- microbes, especially yeasts, have been used from time immemorial for the production of beverages such as wine, beer, whisky, brandy and rum
- the same yeast Saccharomyces cerevisiae used for bread-making, commonly called brewer's yeast, is used for fermenting malted cereals and fruit juices to produce ethanol, which leaves the metabolic reactions behind that production to be recalled
- different types of alcoholic drink are obtained depending on the raw material used and the type of processing, that is with or without distillation
- wine and beer are produced without distillation, whereas whisky, brandy and rum are produced by distillation of the fermented broth

8.4 Antibiotics

What the word means

- antibiotics produced by microbes are regarded as one of the most significant discoveries of the twentieth century and have greatly contributed towards the welfare of human society
- anti is a Greek word meaning against, and bio means life, so together they mean against life, in the context of disease causing organisms, whereas with reference to human beings they are pro life and not against
- antibiotics are chemical substances produced by some microbes that can kill or retard the growth of other, disease-causing, microbes

Penicillin, found by chance

- Penicillin was the first antibiotic to be discovered, and it was a chance discovery
- Alexander Fleming, while working on Staphylococci bacteria, once observed a mould growing in one of his unwashed culture plates around which Staphylococci could not grow, found that this was due to a chemical produced by the mould, and named it Penicillin after the mould Penicillium notatum
- its full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey, and the antibiotic was extensively used to treat American soldiers wounded in World War II
- Fleming, Chain and Florey were awarded the Nobel Prize in 1945 for this discovery

What antibiotics changed

- after Penicillin, other antibiotics were also purified from other microbes
- antibiotics have greatly improved our capacity to treat deadly diseases such as plague, whooping cough or kali khansi, diphtheria or gal ghotu, and leprosy or kusht rog, which used to kill millions all over the globe
- today we cannot imagine a world without antibiotics

8.5 Chemicals, enzymes and other bioactive molecules

Acids, alcohol and enzymes

- microbes are used for the commercial and industrial production of certain chemicals such as organic acids, alcohols and enzymes
- examples of acid producers are Aspergillus niger, a fungus, of citric acid, Acetobacter aceti, a bacterium, of acetic acid, Clostridium butylicum, a bacterium, of butyric acid, and Lactobacillus, a bacterium, of lactic acid, while yeast, Saccharomyces cerevisiae, is used for the commercial production of ethanol
- lipases are used in detergent formulations and are helpful in removing oily stains from the laundry
- bottled fruit juices bought from the market are clearer than those made at home because they are clarified by the use of pectinases and proteases

Three medical molecules

- streptokinase, produced by the bacterium Streptococcus and modified by genetic engineering, is used as a clot buster for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack
- cyclosporin A, used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus Trichoderma polysporum
- statins, produced by the yeast Monascus purpureus, have been commercialised as blood-cholesterol lowering agents, acting by competitively inhibiting the enzyme responsible for the synthesis of cholesterol

8.6 Microbes in sewage treatment

What sewage is

- large quantities of waste water are generated everyday in cities and towns, and a major component of it is human excreta
- this municipal waste-water is also called sewage, and it contains large amounts of organic matter and microbes, many of which are pathogenic
- it cannot be discharged into natural water bodies such as rivers and streams directly, so before disposal sewage is treated in sewage treatment plants (STPs) to make it less polluting, the treatment being carried out by the heterotrophic microbes naturally present in the sewage

Aeration tank, agitated and supplied with air

Flocs, masses of bacteria with fungal filaments

Settling tank, where the flocs sediment

Activated sludge, drawn off from the bottom

Figure 8.F Secondary treatment

8.7 The two stages of treatment

Primary treatment

- these steps basically involve the physical removal of particles, large and small, from the sewage through filtration and sedimentation
- they are removed in stages, floating debris first by sequential filtration, then the grit, that is soil and small pebbles, by sedimentation
- all the solids that settle form the primary sludge and the supernatant forms the effluent, and the effluent from the primary settling tank is taken for secondary treatment

Secondary or biological treatment

- the primary effluent is passed into large aeration tanks, where it is constantly agitated mechanically and air is pumped into it
- this allows vigorous growth of useful aerobic microbes into flocs, masses of bacteria associated with fungal filaments to form mesh like structures, and while growing these microbes consume the major part of the organic matter in the effluent
- this significantly reduces the BOD, the biochemical oxygen demand, of the effluent, BOD being the amount of oxygen that would be consumed if all the organic matter in one liter of water were oxidised by bacteria
- the BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and so measures indirectly the organic matter present in it, and the greater the BOD of waste water the more is its polluting potential

Activated sludge and the digesters

- once the BOD is reduced significantly, the effluent is passed into a settling tank where the bacterial flocs are allowed to sediment, and this sediment is called activated sludge
- a small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum, and the remaining major part is pumped into large tanks called anaerobic sludge digesters
- there, other kinds of bacteria which grow anaerobically digest the bacteria and the fungi in the sludge, producing during that digestion a mixture of gases such as methane, hydrogen sulphide and carbon dioxide, which form biogas and can be used as a source of energy as it is inflammable
- the effluent from the secondary treatment plant is generally released into natural water bodies such as rivers and streams

An aerial view of a sewage plant

Circular tanks laid out across the site

Where the effluent leaves for the river

Figure 8.G An aerial view of a sewage plant

8.8 Sewage treatment at scale

The scale, and where it falls short

- microbes play a major role in treating millions of gallons of waste water everyday across the globe, and this methodology has been practiced for more than a century now in almost all parts of the world, no man-made technology having been able to rival it to date
- due to increasing urbanisation, sewage is being produced in much larger quantities than ever before, yet the number of sewage treatment plants has not increased enough to treat such large quantities
- so untreated sewage is often discharged directly into rivers, leading to their pollution and to an increase in water-borne diseases
- the Ministry of Environment and Forests has initiated the Ganga Action Plan and the Yamuna Action Plan to save these major rivers from pollution, proposing to build a large number of sewage treatment plants so that only treated sewage is discharged into the rivers

8.9 Microbes in the production of biogas

The methanogens

- biogas is a mixture of gases, containing predominantly methane, produced by microbial activity and usable as fuel, and the type of gas produced depends on the microbes and the organic substrates they utilise
- in the fermentation of dough, in cheese making and in the production of beverages the main gas produced was CO_2, but certain bacteria which grow anaerobically on cellulosic material produce a large amount of methane along with CO_2 and H_2
- these bacteria are collectively called methanogens, one common one being Methanobacterium, and they are commonly found in the anaerobic sludge during sewage treatment
- they are also present in the rumen, a part of the stomach, of cattle, where a lot of cellulosic material from the food of cattle is present, and there they help break down cellulose and so matter greatly in the nutrition of cattle, so that the excreta or dung of cattle, commonly called gobar, is rich in these bacteria and can be used to generate biogas, commonly called gobar gas

Concrete tank, 10 to 15 feet deep, holding the slurry

Floating cover, rising as gas collects

Outlet pipe, carrying biogas to nearby houses

Second outlet, from which the spent slurry is removed

Figure 8.H A typical biogas plant

8.10 The biogas plant

How it is built and used

- the plant consists of a concrete tank, 10-15 feet deep, in which bio-wastes are collected and a slurry of dung is fed
- a floating cover is placed over the slurry and keeps rising as gas is produced in the tank by microbial activity, and the plant has an outlet connected to a pipe that supplies biogas to nearby houses
- the spent slurry is removed through another outlet and may be used as fertiliser
- cattle dung is available in large quantities in rural areas where cattle are used for a variety of purposes, so biogas plants are more often built there, and the biogas produced is used for cooking and lighting

Who developed it here

- the technology of biogas production was developed in India mainly due to the efforts of the Indian Agricultural Research Institute (IARI)
- the Khadi and Village Industries Commission (KVIC) shared in that effort
- a visit to a biogas plant near a village, and learning about it from the people actually managing it, is worth making

8.11 Microbes as biocontrol agents

Why not chemicals

- biocontrol refers to the use of biological methods for controlling plant diseases and pests
- in modern society these problems have been tackled increasingly by the use of chemicals, that is insecticides and pesticides
- these chemicals are toxic and extremely harmful to human beings and animals alike, and have been polluting our environment, that is soil and ground water, as well as fruits, vegetables and crop plants, and our soil is also polluted through our use of weedicides to remove weeds

8.12 Biological control of pests and diseases

The organic farmer's view

- in agriculture there is a method of controlling pests that relies on natural predation rather than introduced chemicals, and a key belief of the organic farmer is that biodiversity furthers health, so that the more variety a landscape has the more sustainable it is
- the organic farmer therefore works to create a system where the insects sometimes called pests are not eradicated but kept at manageable levels by a complex system of checks and balances within a living and vibrant ecosystem
- contrary to conventional farming practices, which often use chemical methods to kill both useful and harmful life forms indiscriminately, this is a holistic approach that seeks to develop an understanding of the webs of interaction between the myriad of organisms that constitute the field fauna and flora
- the organic farmer holds that eradicating the creatures often described as pests is not only impossible but also undesirable, since without them the beneficial predatory and parasitic insects which depend on them as food or hosts would not be able to survive

What biocontrol asks of the farmer

- the use of biocontrol measures will greatly reduce our dependence on toxic chemicals and pesticides
- an important part of the biological farming approach is to become familiar with the various life forms that inhabit the field, predators as well as pests
- it also means knowing their life cycles, their patterns of feeding and the habitats they prefer, which helps develop appropriate means of biocontrol

The agents themselves

- the very familiar beetle with red and black markings, the ladybird, and dragonflies are useful to get rid of aphids and mosquitoes respectively
- an example of a microbial biocontrol agent introduced to control butterfly caterpillars is the bacterium Bacillus thuringiensis, often written as Bt, available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where they are eaten by the insect larvae, and in the gut of the larvae the toxin is released and the larvae are killed, while other insects are left unharmed
- because of the development of methods of genetic engineering in the last decade or so, scientists have introduced B. thuringiensis toxin genes into plants, which are then resistant to attack by insect pests, Bt-cotton being one such example, cultivated in some states of our country
- a biological control being developed for use in the treatment of plant disease is the fungus Trichoderma, whose species are free-living fungi very common in root ecosystems and effective biocontrol agents of several plant pathogens

Baculoviruses

- baculoviruses are pathogens that attack insects and other arthropods, the majority of those used as biological control agents being in the genus Nucleopolyhedrovirus
- these viruses are excellent candidates for species-specific, narrow spectrum insecticidal applications, and have been shown to have no negative impacts on plants, mammals, birds, fish or even non-target insects
- this is especially desirable when beneficial insects are being conserved to aid in an overall integrated pest management (IPM) programme, or when an ecologically sensitive area is being treated

8.13 Microbes as biofertilisers

Why the switch

- with our present day life styles environmental pollution is a major cause of concern, and the use of chemical fertilisers to meet the ever-increasing demand of agricultural produce has contributed significantly to it
- there are problems associated with the overuse of chemical fertilisers, and there is a large pressure to switch to organic farming, that is the use of biofertilisers
- biofertilisers are organisms that enrich the nutrient quality of the soil, and their main sources are bacteria, fungi and cyanobacteria

Bacteria, fungi and cyanobacteria

- the nodules on the roots of leguminous plants are formed by the symbiotic association of Rhizobium, and these bacteria fix atmospheric nitrogen into organic forms which the plant uses as nutrient, while other bacteria such as Azospirillum and Azotobacter fix atmospheric nitrogen while free-living in the soil, enriching its nitrogen content
- fungi are also known to form symbiotic associations with plants, that is mycorrhiza, many members of the genus Glomus forming them, and the fungal symbiont absorbs phosphorus from the soil and passes it to the plant
- plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in growth and development, which leaves the advantage to the fungus to be worked out
- cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments, many of which can fix atmospheric nitrogen, such as Anabaena, Nostoc and Oscillatoria, and in paddy fields they serve as an important biofertiliser, blue green algae also adding organic matter to the soil and increasing its fertility

8.14 Summary

Not all microbes are pathogenic

- currently a number of biofertilisers are available commercially in the market, and farmers use these regularly in their fields to replenish soil nutrients and reduce dependence on chemical fertilisers
- microbes are a very important component of life on earth, and many are very useful to human beings, so that we use microbes and microbially derived products almost every day
- bacteria called lactic acid bacteria (LAB) grow in milk to convert it into curd, the dough used to make bread is fermented by the yeast Saccharomyces cerevisiae, certain dishes such as idli and dosa are made from dough fermented by microbes, and bacteria and fungi are used to impart particular texture, taste and flavor to cheese

8.15 Summary, continued

Industry, sewage and biogas

- microbes are used to produce industrial products such as lactic acid, acetic acid and alcohol, which are used in a variety of processes in industry
- antibiotics such as the penicillins, produced by useful microbes, are used to kill disease-causing harmful microbes, and antibiotics have played a major role in controlling infectious diseases such as diphtheria, whooping cough and pneumonia
- for more than a hundred years microbes have been used to treat sewage, that is waste water, by the process of activated sludge formation, and this helps in the recycling of water in nature
- methanogens produce methane, that is biogas, while degrading plant waste, and biogas produced by microbes is used as a source of energy in rural areas

Biocontrol and biofertilisers

- microbes can also be used to kill harmful pests, a process called biocontrol
- biocontrol measures help us avoid the heavy use of toxic pesticides for controlling pests
- there is a need these days to push for the use of biofertilisers in place of chemical fertilisers
- it is clear from the diverse uses human beings have put microbes to that they play an important role in the welfare of human society

8.16 Exercises

From this chapter

1 Bacteria cannot be seen with the naked eyes, but these can be seen with the help of a microscope. If you have to carry a sample from your home to your biology laboratory to demonstrate the presence of microbes with the help of a microscope, which sample would you carry and why?
2 Give examples to prove that microbes release gases during metabolism.
3 In which food would you find lactic acid bacteria? Mention some of their useful applications.
4 Name some traditional Indian foods made of wheat, rice and Bengal gram (or their products) which involve use of microbes.
5 In which way have microbes played a major role in controlling diseases caused by harmful bacteria?
6 Name any two species of fungus, which are used in the production of the antibiotics.
7 What is sewage? In which way can sewage be harmful to us?
8 What is the key difference between primary and secondary sewage treatment?
9 Do you think microbes can also be used as source of energy? If yes, how?
10 Microbes can be used to decrease the use of chemical fertilisers and pesticides. Explain how this can be accomplished.
11 Three water samples namely river water, untreated sewage water and secondary effluent discharged from a sewage treatment plant were subjected to BOD test. The samples were labelled A, B and C; but the laboratory attendant did not note which was which. The BOD values of the three samples A, B and C were recorded as 20mg/L, 8mg/L and 400mg/L, respectively. Which sample of the water is most polluted? Can you assign the correct label to each assuming the river water is relatively clean?

8.17 Exercises, continued

From this chapter, continued

1. Find out the name of the microbes from which Cyclosporin A (an immunosuppressive drug) and Statins (blood cholesterol lowering agents) are obtained.
2. Find out the role of microbes in the following and discuss it with your teacher. (a) Single cell protein (SCP) (b) Soil
3. Arrange the following in the decreasing order (most important first) of their importance, for the welfare of human society. Give reasons for your answer. Biogas, Citric acid, Penicillin and Curd
4. How do biofertilisers enrich the fertility of the soil?